Extending the PISCO gridded dew-point record for Peru (2017–2020) from the continued minimum-temperature record, using per-climate-zone seasonal model selection

Piero Palacios
International Potato Center (CIP)
piero.palacios@cgiar.org

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> ! Important
>
> **DRAFT: not for circulation.** All quantitative results below are AI-assisted analyses that **require subject-matter-expert (SME) validation before use** (CIP Acceptable Use policy; see Section 9 and Section 10). Open items are marked **[SME]** or **[TODO]**.

Abstract

The PISCO gridded climate product for Peru provides daily dew-point temperature (Td) only through 2016, while its minimum-temperature (Tmin) record continues to 2020. This leaves a four-year gap in a variable required for agricultural and hydrological modelling. We extend the Td record to 2017–2020 by learning a Tmin⟶Td mapping on the 1981–2016 overlap and applying it forward at every grid cell. The model is a per-(cell, day-of-year) weighted local linear regression on climate anomalies, in which both the predictor set and the day-of-year smoothing half-window are selected per **SENAMHI climate zone × day-of-year** by leave-one-year-out cross-validation, following the MultiLLR approach of J. Hwang, P. Orenstein, J. Cohen, K. Pfeiffer, and L. Mackey [1]. Because no observed Td exists in the gap, prediction is **autoregressive**: each predicted day feeds the following day's Td lags.

Validation is **temporal, not spatial**: we hold out the most recent four years (train 1981–2012, predict 2013–2016) and score against observed Td under the same autoregressive conditions the real gap imposes. The tuned model attains RMSE 1.219 °C versus 1.535 °C for climatology (the incumbent gap-filling choice), a reduction of 0.316 °C [95% CI 0.314, 0.317], skill 0.206, outperforming climatology at 99.99% of grid cells (Cohen's $d = 2.98$, permutation $p \simeq 2 \times 10^4$). It also beats Td = Tmin (FAO-56) by 2.33 °C and a per-zone OLS fit by 1.11 °C. Re-selecting recipes on training years only changes skill by 0.009, confirming the result is not an artefact of selection.

The resulting product covers 2,178,827 grid cells × 1,461 days (3.18×10^9 values). It shows no discontinuity at the 2016/2017 boundary: over equal-length four-year windows the filled field differs from the last observed window (mean −0.041 °C, sd 0.176 °C) no more than two observed windows differ from each other (mean +0.101 °C, sd 0.125 °C), with no grid cell exceeding 1 °C in either comparison. A known limitation is a small warm tail (0.13% of values > 35 °C) confined to the north-eastern Amazon lowlands and concentrated almost entirely in **November**, the dry-to-wet transition (1.56% of November values versus 0.03% in October). Diagnostics over all 3.18×10^9 values rule out autoregressive error accumulation and localise the cause to variance inflation where model selection shrinks the day-of-year smoothing window at the dry-to-wet transition, affecting the tail of the predictive distribution while leaving mean skill unchanged. **[SME]**

1 Introduction

Dew-point temperature is a primary input to reference-evapotranspiration, disease-risk and crop-growth models. For Peru, the PISCO family of gridded products [2] provides the national-scale daily fields these applications need. However, the products are versioned and released asynchronously: dew point is available from the v1.1 release for **1981–2016**, whereas minimum

temperature has been extended in v1.2 to **1981–2020**. Applications requiring Td after 2016 therefore face a four-year gap.

The standard fallbacks are unattractive. Substituting the **daily climatology** discards all inter-annual and synoptic information. The FAO-56 approximation **Td ≈ Tmin** [3] is convenient but biased in arid and high-altitude terrain. A single national regression of Td on Tmin ignores the extreme heterogeneity of Peruvian microclimates, which span hyper-arid coast, high Andes and humid Amazon within a few hundred kilometres.

We therefore ask a narrow, answerable question: **can the Tmin→Td relationship learned on the 1981–2016 overlap reconstruct Td for 2017–2020 more accurately than these incumbents?** Two features distinguish this from a conventional spatial-interpolation problem:

1. **It predicts new *years* at the *same* grid cells**, not new locations. Validation must therefore be temporal (hold out years), not spatial (hold out sites).
2. **No observed Td exists in the gap.** Because the model uses Td lags as predictors, prediction must be autoregressive, and any honest validation must be run the same way.

2 Data

Table 1: Data inventory.

Item	Source	Coverage	Role
Dew point Td	PISCO **v1.1**	1981–2016	Target (the record being extended)
Minimum temperature Tmin	PISCO **v1.2**	1981–2020	Predictor (train **and** predict)
Grid	PISCO national grid	2,818,404 cells	Model domain
Climate zones	SENAMHI Thornthwaite classification	41 zones	Unit of model selection

Version consistency (important). Tmin v1.1 and v1.2 differ enough that mixing them injects a systematic bias: the fitted Tmin sensitivity differs by version (coefficient mean 0.45 vs 0.37) and switching versions between training and prediction shifts Td by ≈0.3 °C. We therefore use **v1.2 Tmin on both sides**, for training and prediction, so the mapping is applied to the same data generating process it was learned from.

Grid alignment (important). The raw v1.2 Tmin distribution is supplied on a 302,449-point subset grid whose cell indices do **not** correspond to those of the 2,818,404-cell national grid used by the Td product. Applying it directly produces physically meaningless results (we measured a spurious −8.7 °C step at the 2016/2017 boundary). All future Tmin was therefore re-extracted onto the full national grid; we verified the resulting grid index is byte-identical to that of the Td product.

Of the 2,818,404 cells, **2,178,827** (77.3%) yield predictions; the remainder are ocean or otherwise lack the climatology or Tmin required by the model.

Caveat on the target. Validation is against the *gridded* v1.1 Td, the product being extended, which is the appropriate reference for this task. Gridded Td and Tmin both derive from land surface temperature and elevation covariates in the PISCOeo_pm framework [2], so they share some structure; Td is not derived from Tmin directly. Station-level validation is out of scope (Section 7).

3 Methods

3.1 Anomaly formulation

All modelling is performed on **anomalies** relative to a per-(cell, day-of-year) daily climatology computed over the training years:

$$X_{\text{anom}}(t) = X(t) - \bar{X}(\text{doy}(t))$$

For each cell and day-of-year the model is a weighted local linear regression

$$\text{Td}_{\text{anom}}(t) = \beta_0 + \sum_k \beta_k \, x_k(t) + \varepsilon$$

where candidate predictors x_k are drawn from the pool $\{\text{TMIN_anom}, \text{TMIN_anom_lag}_{1,2,7,30}, \text{TD_anom_lag}_{1,2,3,7,30}\}$ plus an intercept (11 candidates). Td enters only through lags, never contemporaneously. Fitting uses a circular tricube kernel over day-of-year with half-window h, so each (cell, doy) model borrows strength from nearby days of year across all training years.

3.2 Per-zone × day-of-year model selection

Rather than fixing the predictor set and h globally, we select both **per SENAMHI climate zone × day-of-year** (41 × 366 = 15,006 recipes), following the multitask MultiLLR design of J. Hwang, P. Orenstein, J. Cohen, K. Pfeiffer, and L. Mackey [1]: predictors are chosen jointly across grid points within a zone, while coefficients are fitted per grid point. Selection is by **backward-stepwise elimination** under a **leave-one-year-out cross-validated (LOYOCV) cosine-skill** objective, with h searched over $\{7, 11, 15, 21, 31, 45\}$ days. The approach has since been benchmarked against other machine-learning methods for subseasonal forecasting [4] and applied to precipitation forecasting elsewhere [5]; we adapt it here to a reconstruction rather than a forecasting task.

Selection runs on a stratified sample of **66,534 cells** (target 2,000 per zone), which is sufficient to *choose* a recipe, since LOYOCV skill sums are additive across cell chunks, rather than on the full grid.

3.3 Training

The selected recipes are then applied to fit coefficients at **all 2,818,404 grid cells** in a single pass, each (cell, doy) using its zone's recipe as a feature mask. This yields 8.13×10^9 tidy coefficient rows.

3.4 Autoregressive forecast

For 2017–2020 no observed Td exists, so `TD_anom_lag`$_k$ predictors must be supplied by the model's own earlier predictions. The forecast is seeded with observed Td from December 2016 and advances day by day: the value predicted on day t becomes an input for days $t+1 \ldots t+k$. Tmin

predictors are exogenous and observed throughout, so they require no recursion. This sequential dependence is intrinsic to the method, not an implementation limitation.

3.5 Implementation

Selection and training are GPU-accelerated (CuPy, single NVIDIA RTX A2000 12 GB); the grid is partitioned into 8,192 buckets so per-bucket tensors fit in device memory. Full-grid selection plus training took ≈39.5 h wall-clock. The production fill parallelises across buckets on 14 CPU processes and completes in ≈20 min. Code: `HPC_code_tunning/` (see Section 8).

4 Validation design

The evaluation must answer: *how accurate will the 2017–2020 fill be?* Since that answer cannot be observed, we reproduce its conditions on years we can score.

- **Forward temporal hold-out.** Train on early years, predict the most recent years, score against observed Td. The headline split is **leave-last-4** (train 1981–2012, predict 2013–2016), mirroring the real four-year gap; leave-last-1 and leave-last-2 give the lead-time curve.
- **Autoregressive throughout.** No observed Td enters the held-out block, exactly as in production.
- **No leakage.** Climatology, used both as a baseline and as the anomaly anchor, is recomputed on training years only.
- **Baseline ladder**, scored on the identical hold-out: climatology, Td = Tmin [3], per-zone OLS Td~Tmin, and the tuned model. We use the **uncorrected** FAO-56 form; established aridity-dependent corrections exist [6], [7] and a −2 °C offset is common practice in arid and semi-arid settings [8]. Over our domain the uncorrected form is the *more favourable* variant: its measured bias is −2.13 °C (it under-predicts observed Td), so a further −2 °C offset would increase the error. However, a regionally applied correction (coast and Andes only) would be the fairer comparison and is left for future work. **[TODO]**
- **Sample.** 64 whole buckets = 17,041 cells. Because bucket assignment is `ID mod 8192`, each bucket is a nationwide lattice spanning all 41 zones, making a few buckets a cheap, representative sample.
- **Effect sizes over p-values.** ΔRMSE in °C with location-block bootstrap 95% CIs, skill score, fraction of cells improved, Cohen's d, and a sign-flip permutation null (5,000 draws).

Selection honesty. In the primary run the *recipe* came from the full-period manifest, which had seen the hold-out. We therefore repeated the exercise re-selecting recipes on training years only; the difference bounds the selection optimism (Section 5.2).

5 Results

5.1 Model selection

Selection retains most candidates: the full 11-predictor set is kept for 70.7% of zone-days, and no predictor is retained in fewer than 91% of recipes. This indicates that the conventional fixed five-predictor configuration is *under*-specified. The chosen half-window distributes across the whole grid (h = 7: 14.2%, 11: 9.2%, 15: 9.4%, 21: 11.1%, 31: 13.0%, 45: 43.1%). That $h = 45$ is modal is itself

a finding: the model prefers broad day-of-year pooling, trading seasonal specificity for sample size. Extending the grid beyond 21 days was worthwhile but second-order (hold-out skill 0.197 $\longrightarrow$ 0.200).

In-sample LOYOCV uplift over the fixed five-predictor baseline is small (mean +0.015, median +0.008 skill; 16.8% of zone-days show no gain). **These in-sample numbers are selection-optimistic and are not the headline**; Section 5.2 is.

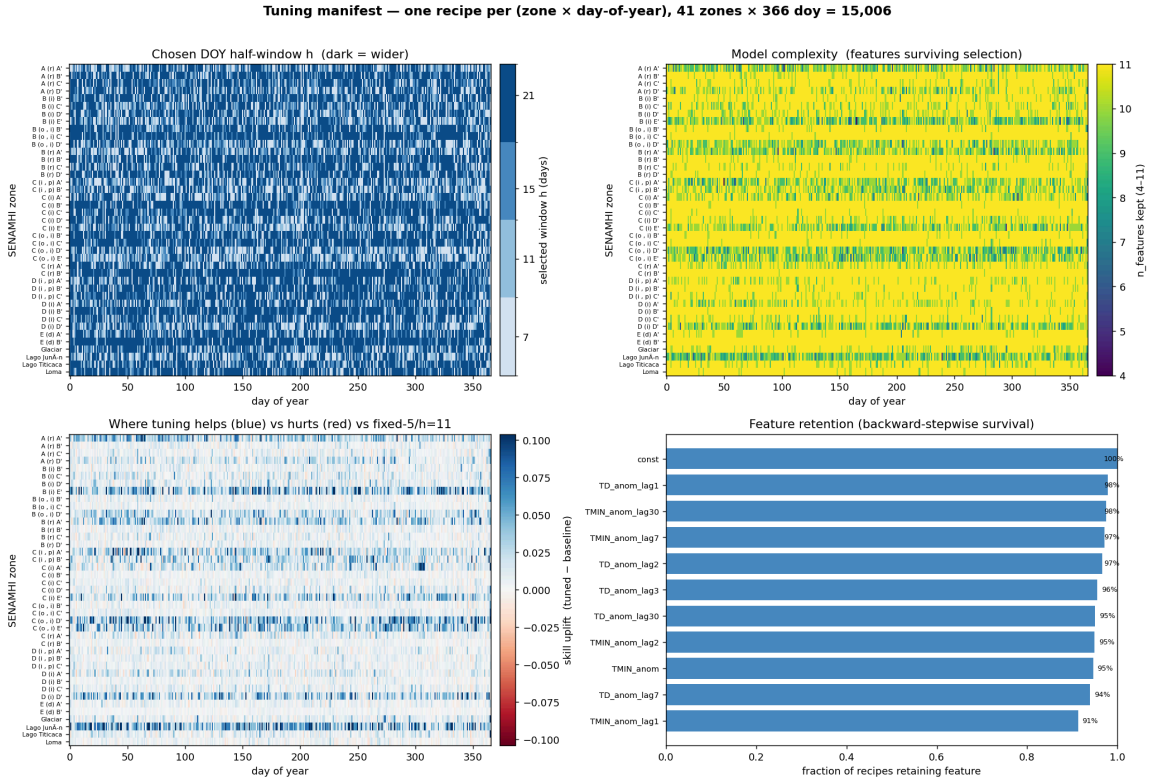


Figure 1: Model-selection diagnostics: distribution of the selected day-of-year half-window h, predictor retention, and LOYOCV skill uplift over the fixed-5 baseline.

5.2 Temporal validation

The tuned model outperforms every incumbent at every lead time (Table 2, Figure 2).

Table 2: Hold-out RMSE (°C), pooled over all held-out cell-days (ll4: n = 24,896,901).

Split (train → predict)	Climatology	Td = Tmin	Zone OLS	**Tuned**	Skill vs clim.
ll1 (1981–2015 → 2016)	1.680	3.621	2.379	**1.262**	**0.249**
ll2 (1981–2014 → 2015–16)	1.667	3.676	2.370	**1.274**	**0.235**
ll4 (1981–2012 → 2013–16)	**1.535**	**3.551**	**2.329**	**1.219**	**0.206**

For the headline ll4 split, paired per-cell effect sizes (17,041 cells):

Table 3: Effect sizes, ll4 split. CIs from location-block bootstrap.

Tuned vs	ΔRMSE °C [95% CI]	Skill	Cells improved	Cohen's d	perm. p
Climatology	+0.316 [0.314, 0.317]	0.206	99.99%	2.98	2×10^{-4}
Td = Tmin	+2.332 [2.308, 2.354]	0.657	99.98%	1.37	2×10^{-4}
Zone OLS	+1.110 [1.091, 1.126]	0.476	99.80%	1.06	2×10^{-4}

The tuned model shows bias −0.046 °C and r = 0.990. Skill declines monotonically with lead time (0.249 → 0.235 → 0.206), the expected growth of error under autoregression, without collapsing over a four-year horizon (Figure 3).

Selection bias is negligible. Re-selecting recipes on training years only lowers ll4 skill from 0.206 to 0.197, a difference of 0.009 skill (≈0.013 °C RMSE, ~4% of the uplift). The advantage over climatology is therefore a property of the method, not of having seen the hold-out.

Seasonal bias is corrected. The fixed-configuration model carries a +1.45 °C December warm bias; under the per-zone × day-of-year recipes the December bias falls to +0.01 °C, with all months within ±0.44 °C.

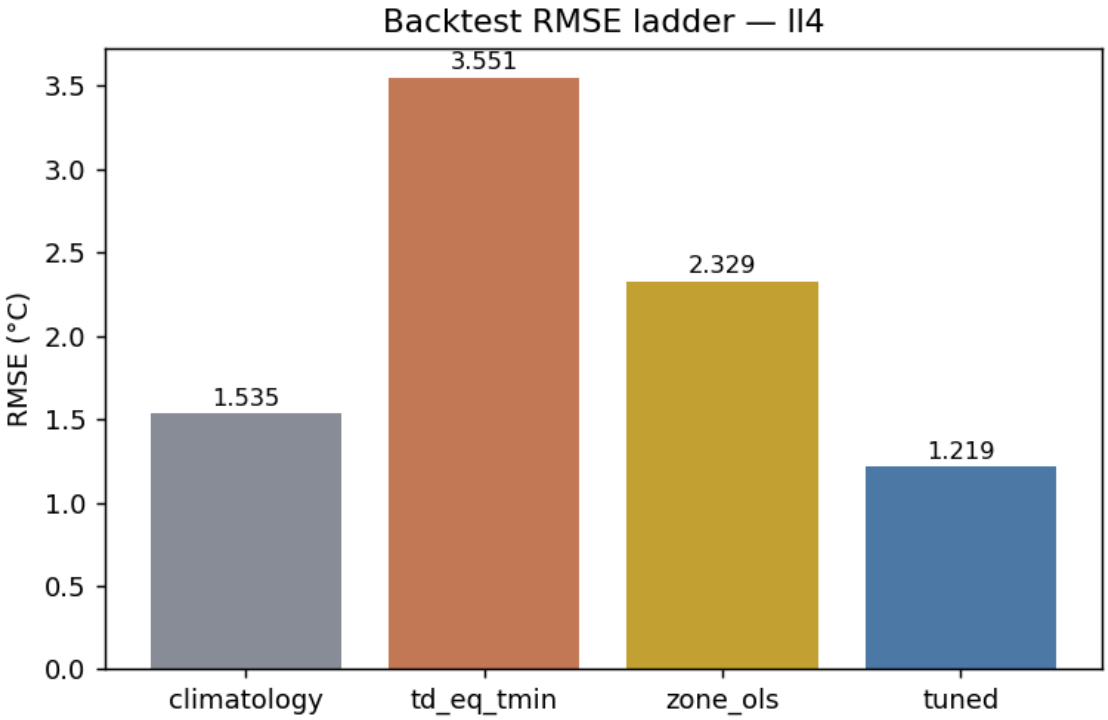


Figure 2: Hold-out RMSE ladder, ll4 split: the tuned model against the three incumbents.

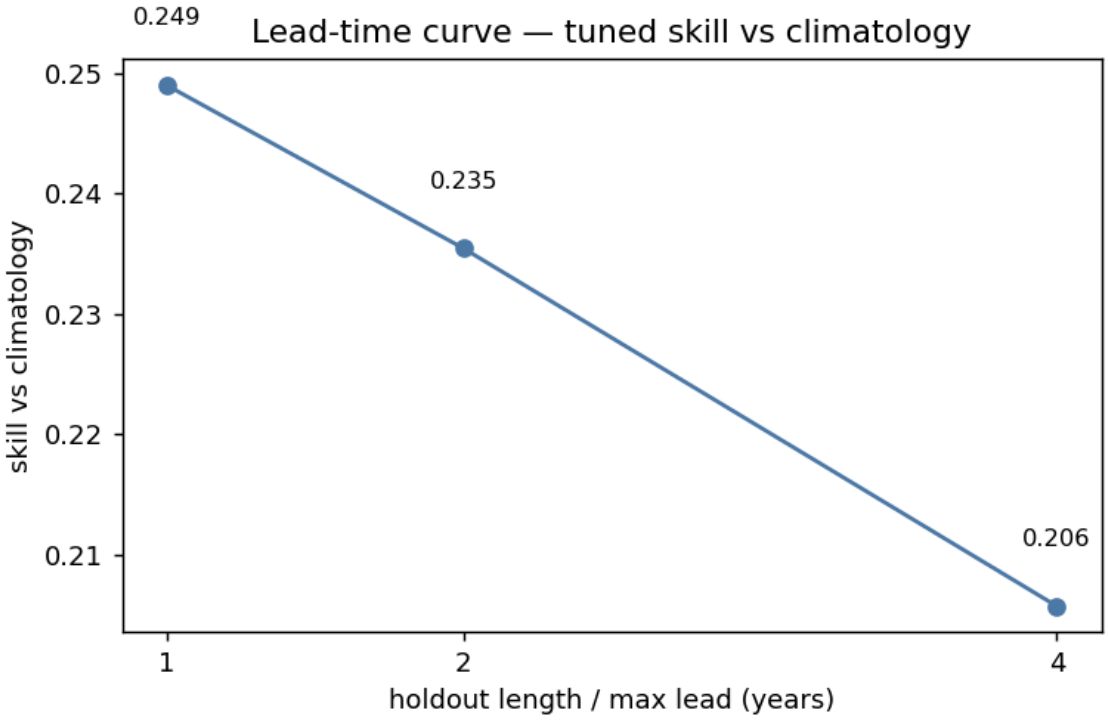


Figure 3: Skill versus climatology as a function of forecast lead time.

5.3 The gap-filled product

Applying the coefficients trained on all of 1981–2016 to the observed 2017–2020 Tmin yields **3,183,266,247** predicted values over 2,178,827 cells and 1,461 days, with no missing values. National mean Td is 19.39 °C (range −13.29 to 41.16 °C); annual means are stable across the horizon (19.36, 19.30, 19.41, 19.51 °C for 2017–2020), showing no drift under sustained autoregression.

The filled field reproduces the expected national structure, with a humid Amazon (~25 °C), a sharply delineated cold Andean cordillera, and the coastal gradient (Figure 5), and it continues the observed seasonal cycle across the 2016/2017 boundary with matching amplitude and phase (Figure 4).

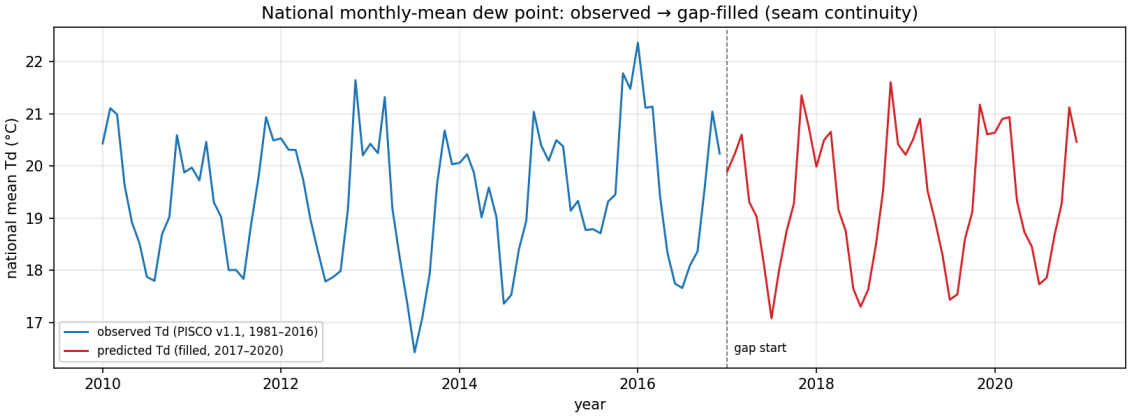


Figure 4: National monthly-mean Td, 2010–2020: observed (blue, through 2016) and gap-filled (red, 2017–2020).

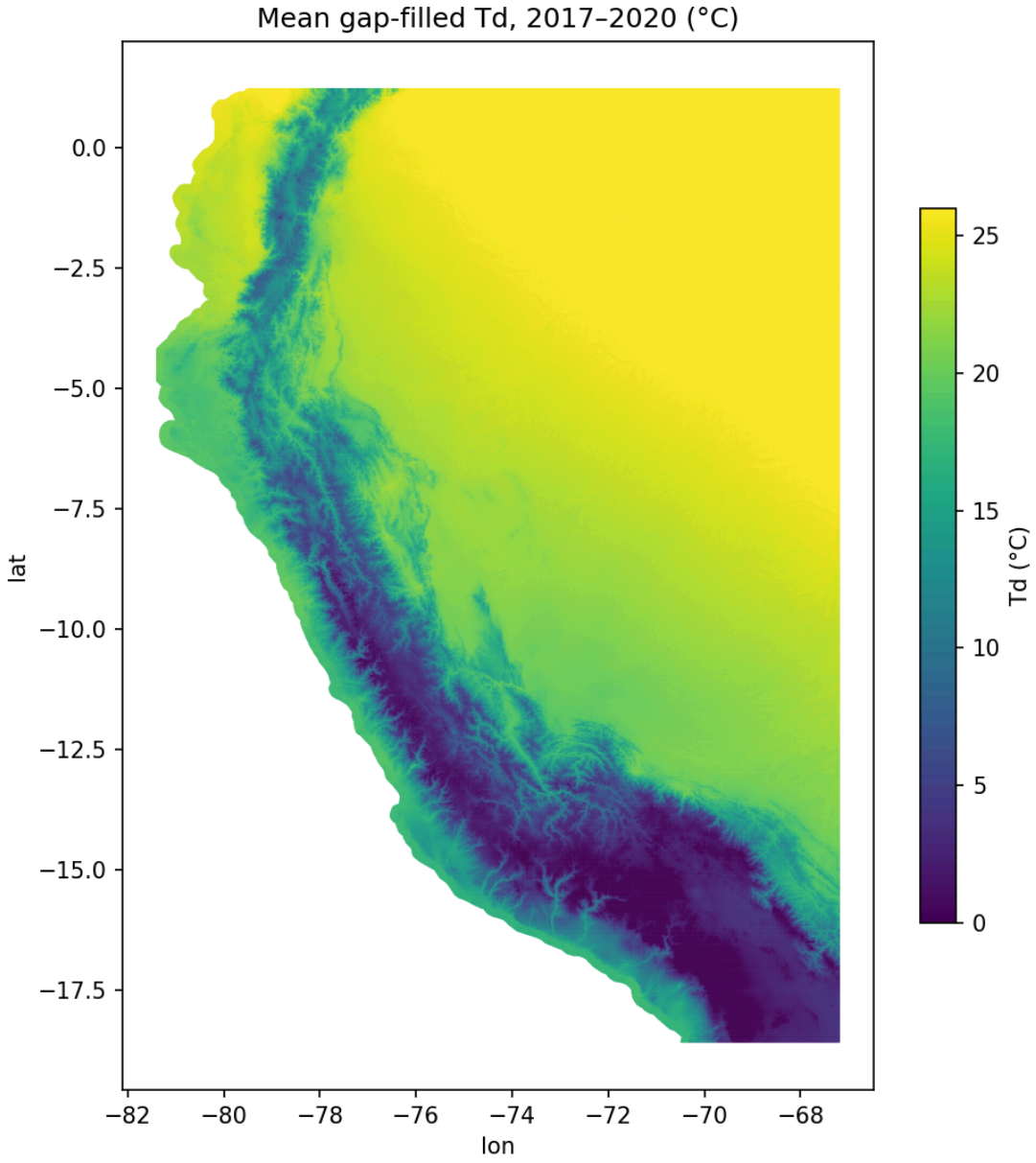


Figure 5: Mean gap-filled Td, 2017–2020.

5.4 Seam continuity

A gap-filled record must not introduce an artificial step where prediction begins. Comparing adjacent *months* across the boundary conflates the real seasonal cycle with any such step, so we test on **equal-length full-year windows**, which removes seasonality, and include a **model-free**

control: the difference between two consecutive observed four-year windows, which measures the natural period-to-period variability of the observed record itself.

Table 4: Difference in annual-mean Td (°C), 2,178,827 cells.

Comparison	Mean	sd	1st–99th pct	Cells $\|\Delta\| > 1$ °C
A obs 2013–16 − obs 2009–12 *(control, no model)*	+0.101	0.125	−0.18 … +0.42	0.0%
B pred 2017–20 − obs 2013–16 *(seam)*	−0.041	0.176	−0.36 … +0.54	0.0%

The filled years differ from the last observed window no more than two observed windows differ from each other: B's spread is 1.4× A's, its mean is closer to zero, and **no cell in either comparison differs by more than 1 °C**. Panel B is spatially smooth and coherent, without the blockiness or speckle that a numerical artefact would produce (Figure 6).

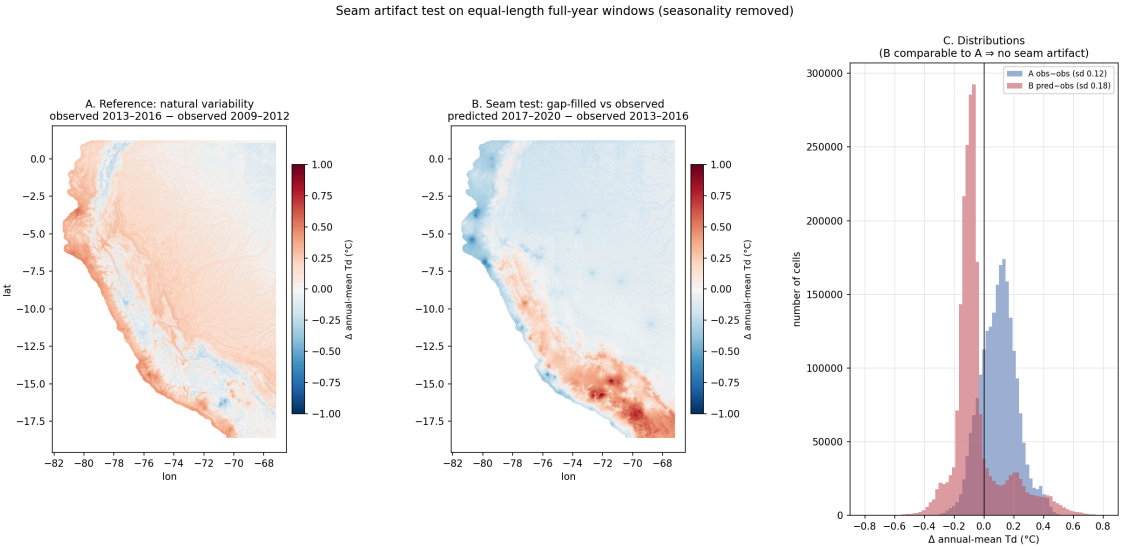


Figure 6: Seam artefact test on equal-length full-year windows. **A** natural variability (observed only); **B** the seam; **C** distributions.

6 Discussion

6.1 Relation to existing humidity-from-temperature methods

The FAO-56 identity Td ≈ Tmin [3] is a starting point rather than the state of the art. Its bias in arid, semi-arid and high-elevation terrain is well documented, and several corrections are in routine use: an aridity-dependent adjustment driven by precipitation and potential evapotranspiration [6], a correction from temperature and precipitation validated across alpine stations [7], and a

simple −2 °C offset widely applied in arid and semi-arid regions [8]. Disaggregation frameworks such as MELODIST implement both the naive identity and fitted regression alternatives precisely because the identity fails in a site- and regime-specific way [9].

The most directly relevant prior result is D. Althoff and L. Rodrigues [10], who calibrate the temperature-based Penman-Monteith approach over the Brazilian Cerrado, a tropical biome with a pronounced dry season, and find **corrections of opposite sign in the two seasons**: −0.93 °C in the wet season (Td > Tmin) versus +0.57 °C in the dry season (Td < Tmin). The Tmin–Td coupling is therefore not a fixed offset but a *regime-dependent* relationship. Our per-(zone × day-of-year) design can be read as a generalisation of that finding: rather than two seasonal offsets for one biome, we fit a separate local regression for each climate zone and each day of the year. The magnitude of our improvement over both Td = Tmin (+2.33 °C RMSE) and a per-zone OLS fit (+1.11 °C) is consistent with this interpretation: a single global or even per-zone linear relationship is too rigid.

6.2 What the selected smoothing windows imply

Selection favours wide day-of-year windows, with $h = 45$ days modal at 43.1% (Figure 1). An independent precedent exists: G. Chabot-Couture, K. R. Nigmatulina, and P. Eckhoff [11] interpolate dew point with day-of-year-dependent parameters smoothed over **31-day** windows, a comparable scale. The physical reading is that the Tmin–Td relationship varies smoothly with day of year, so pooling neighbouring days buys sample size at little cost in bias; the model prefers more data over finer seasonal resolution.

That preference is not uniform, however, and the national modal value conceals substantial between-zone variation: the humid-rainforest zone that dominates the Amazon lowlands prefers markedly narrower windows (median $h = 21$, only 31.7% at $h = 45$), and narrows them further still at the dry-to-wet transition. We return to this in Section 7, because that local narrowing, not the national preference for broad pooling, turns out to drive our one substantive failure mode.

6.3 Positioning relative to the gap-filling literature

Most work described as "gap filling" reconstructs values **missing inside** an existing record (station outages, sensor dropouts, cloud-obscured retrievals), using the surrounding observations in space and time [12]. Our problem is different in kind: **temporal extrapolation beyond the end of the record**, where the target variable is unobserved *everywhere* throughout the prediction window. No contemporaneous Td exists anywhere in 2017–2020 to interpolate from.

This distinction has a direct methodological consequence. Spatial cross-validation, the prevailing validation strategy, for instance the spatial leave-one-out used in global spatio-temporal regression kriging of daily temperature [13], cannot measure the quantity we care about, because it holds out *locations* while retaining contemporaneous observations. Only a forward temporal hold-out reproduces the operational situation. Explicit comparisons of the two designs, and systematic reporting of incumbent baselines such as climatology and persistence, remain uncommon in this literature. We therefore treat the temporal hold-out as primary and report the full baseline ladder with effect sizes and confidence intervals (Section 5.2).

6.4 Autoregressive stability

Because Td lags are predictors and no observed Td exists in the gap, the forecast is a 1,461-step recursion. The deep-learning climate literature documents that such rollouts can degrade progressively or diverge outright through error accumulation, and that explicit multi-step training is needed to stabilise them [14]. We see no comparable degradation: skill decays gracefully with lead time ($0.249 \longrightarrow 0.235 \longrightarrow 0.206$ from one to four years, Figure 3), annual mean Td is stable to within 0.21 °C across the four filled years, and the incidence of extreme values is **non-monotonic** in time rather than growing (Section 7).

Two features of the design likely account for this. First, the dominant predictors are anomalies of **observed, exogenous** Tmin, which are supplied from data at every step and never recurse; only the Td-lag channel is autoregressive. Second, predictions are formed as anomalies added back to a fixed per-cell climatology, which anchors the recursion to a physically sensible level. Free-running emulators that must propagate their entire state forward have neither property. This suggests a general point: **hybrid recursions with observed exogenous forcing and a climatological anchor are substantially more stable over long horizons than fully autoregressive ones**, and the stability concerns of that literature do not transfer directly to this class of model.

7 Limitations

Warm tail. [SME] 0.1326% of predicted values exceed 35 °C (maximum 41.16 °C), which is not physically plausible for dew point. We characterised this failure mode over all 3.18×10^9 values; the diagnostics constrain its cause more tightly than the raw rate suggests.

Where and when. The affected cells (132,920 with at least one exceedance) lie in the north-eastern Amazon lowlands, and the exceedances are concentrated almost entirely in a **single month, November**, in every year: 2.41% of values in November 2017, 1.91% in November 2018, 1.55% in November 2020 and 0.36% in November 2019, against 0.081% for the next-worst month (October 2018) and below 0.03% elsewhere. Pooling the four years by month of year sharpens the picture further (Figure 7): **1.556% in November against 0.027% in October, a factor of 58, and only 50 exceedances in total across January–July** (out of $\approx 1.9 \times 10^9$ values in those months). November corresponds to the dry-to-wet transition in this region **[SME: confirm regional seasonality]**.

Not autoregressive drift. Annual incidence is **non-monotonic** (0.2015%, 0.1696%, 0.0293% and 0.1299% for 2017–2020), and exceedances occur in short episodes: 40.6% are isolated single days, mean run length 3.42 days, longest 36 days. Progressive error accumulation of the kind reported for free-running autoregressive models [14] would produce growth with lead time and lengthening runs; neither appears. This is consistent with the exogenous-forcing and climatological-anchoring argument in Section 6.

Why a single month: diagnosis. We ran three internal diagnostics to discriminate between candidate explanations. They rule out our initial hypothesis and support a different one.

Where. Every affected cell lies in a *single* SENAMHI zone: zone 0, Thornthwaite class `A (r) A'` (perhumid, no water deficiency, megathermal), the humid tropical rainforest zone, comprising 132,920 of that zone's 909,708 cells (14.6%). No other zone is affected.

The failure is in the variance, not the mean. November's leave-one-year-out skill in this zone (0.7317) is indistinguishable from December (0.7333) and January–July (0.7324), and *better* than August–September (0.7030); in-sample R^2 shows the same pattern (0.605 in November against 0.641 in December, 0.590 in August–September). November is therefore **not** anomalously badly fitted on average, which is consistent with annual means remaining stable to 0.21 °C. Whatever goes wrong affects the tail of the predictive distribution, not its centre.

The selected windows in November are narrow, not wide. This overturns our initial reading. While $h = 45$ is the modal choice nationally (43.1%), that statistic does not hold in the affected zone: zone 0 prefers narrower windows overall (median $h = 21$; only 31.7% at $h = 45$), and **November is its narrowest band**: 47% of November days select $h \leq 11$, against 23% in October, with median $h = 15$ in November versus 31 in *both* adjacent months. Selection is evidently responding to a rapidly changing time of year by shrinking the smoothing window to track it.

That resolves the puzzle, and it is a textbook bias–variance trade-off. Narrowing the window reduces bias where the underlying relationship changes quickly, but sharply reduces the effective sample behind each per-(cell, day-of-year) fit, inflating coefficient variance. Because our failure metric is an **exceedance rate above a fixed threshold**, variance rather than bias dominates it: a modest increase in predictive spread, invisible in mean skill, moves the >35 °C rate by a factor of 58. It also explains the sharpness directly, since narrow windows are selected *only* in the transition band, so the variance inflation is confined there rather than smeared across ±45 days.

The physical transition remains the trigger. The Amazonian dry-to-wet transition is a genuine rapid regime shift, initiated by late-dry-season rainforest transpiration activating shallow convection [15], driven by latent-heat-flux increases that rapidly reduce convective inhibition [16], with an accompanying circulation reorganisation [17], and routinely dated to a specific pentad [18]. That is plausibly *why* selection narrows the window there. It is also the classical setting in which nonparametric smoothing struggles, since such methods "require the crucial assumption that the underlying trend function is smooth" [19].

The exceedance peak moves between years, as an onset-linked mechanism predicts and a fixed calendar defect would not: the peak day-of-year is 313 (9 Nov) in 2017, 331 (27 Nov) in 2018, 313 in 2019 and 331 (26 Nov) in 2020, with the distribution mean shifting between day 314.9 and 322.2. Onset timing is itself modulated non-linearly by Pacific and Atlantic sea-surface temperatures and by the MJO [18], [20], which offers a route to explaining the sevenfold interannual range in severity, though we have not tested against onset indices directly. **[TODO]**

A published precedent, and a non-stationary target. This class of artefact is not new: G. Chabot-Couture, K. R. Nigmatulina, and P. Eckhoff [11] report interpolated dew point exceeding interpolated air temperature (implying RH > 100%) in a day-of-year-parameterised dew-point interpolation, and recommend capping. The affected regime is also changing over time, which compounds the difficulty of fitting it from a 1981–2016 sample: dry-season vapour pressure deficit over the Amazon has been rising [21] and vertical drying has been observed over deforested Amazonia [22]. We did not test for non-stationarity in our training window. **[TODO]**

Remedy implied by the diagnosis. Because the mechanism is variance inflation from window shrinkage, the targeted fix is to **constrain the minimum window** (or otherwise regularise the fits) for the transition band in this zone, rather than to clip outputs after the fact. This is untested and is not applied to the product. **[TODO]**

A cap at Td ≤ Tmin is not viable. We tested it: 79.6% of all predicted values exceed the same-day Tmin, by a mean of 3.0 °C, and so does the *observed* gridded Td, since the Td = Tmin baseline has a bias of −2.13 °C over this domain. Capping at Tmin would therefore alter 79.6% of the product with a mean shift of 2.38 °C, which is not a correction but a distortion. The warm-tail values are nonetheless extreme on this measure too: all of them exceed same-day Tmin, by a mean of 11.4 °C against the 3.0 °C typical excess. A bound on the *excess* (Td ≤ Tmin + k) or on daily maximum temperature is therefore worth evaluating, as is refitting the affected zones with explicit wet/dry stratification. **[TODO]**

Open question. The interannual spread (November 2019 roughly seven times weaker than November 2017) points to modulation by interannual climate variability, plausibly through its control on onset timing [20]; testing this against onset-date indices is left for future work. **[TODO]**

No correction is applied in the product as it stands (Figure 8, Figure 7).

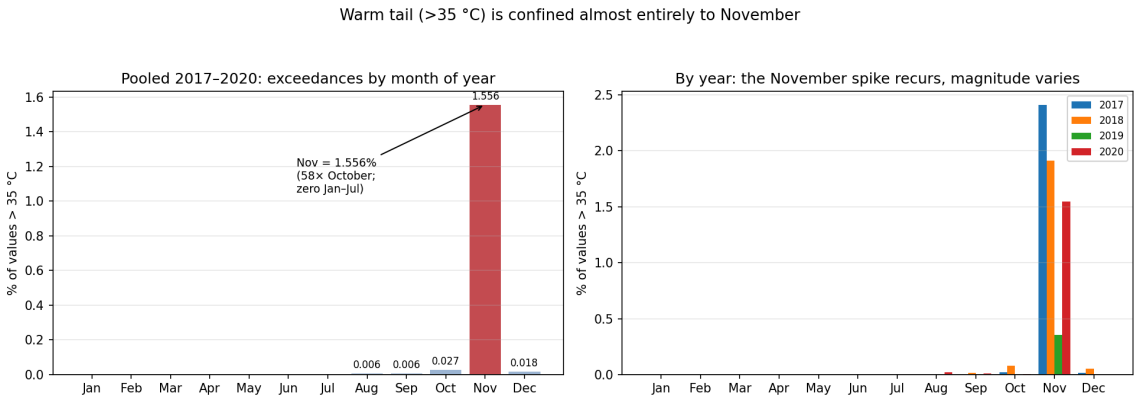


Figure 7: Warm-tail exceedances by month of year. Left: pooled 2017–2020. Right: by year, showing that the November spike recurs annually while its magnitude varies by a factor of seven.

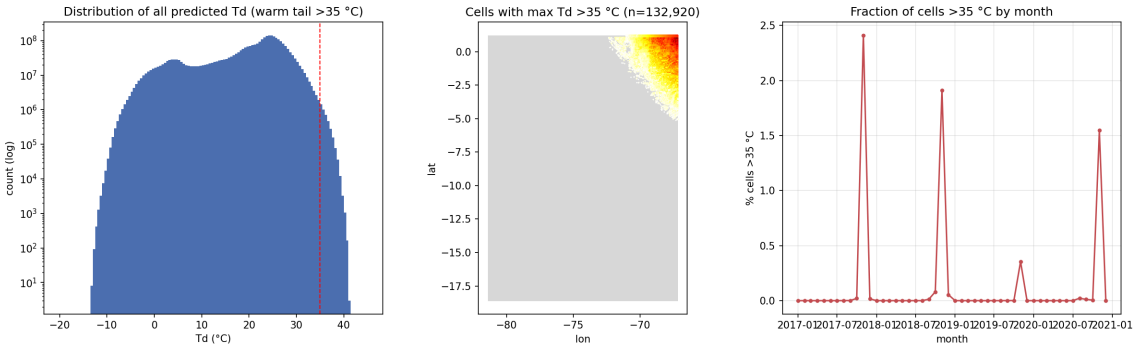


Figure 8: Warm tail: value distribution, spatial location of affected cells, and monthly incidence.

Regional pattern at the seam. Panel B of Figure 6 shows a coherent regional signal (warmer southern Andes, cooler northern Amazon). Because no observed Td exists after 2016, that figure alone cannot separate a genuine 2017–2020 climate anomaly from regional model bias. The bound comes from Section 5.2, where regional errors were measured directly on held-out years. **[SME]**

Shared covariates. Gridded Td and Tmin share LST and elevation predictors in their construction, so validation against gridded Td may flatter the mapping relative to an independent reference.

No station-level validation. The target is the gridded product being extended, which is the correct reference for this task, but it means the result inherits any bias of that product. Station-level evaluation is out of scope here and is the natural next step. **[TODO]**

Autoregressive horizon. Skill decays with lead time; 2020 predictions rest on a longer recursive chain than 2017. The lead-time curve quantifies this to four years, the length of the actual gap.

Faint circular features visible in Figure 6 appear in the observed-only control panel as well and are therefore attributable to station influence in the underlying PISCO regression-kriging, not to this method.

8 Data and code availability

- **Code.** `HPC_code_tunning/`: selection (`selection.py`), training (`train_zoned.py`), autoregressive forecast (`forecast_zoned.py`), production fill (`fill_zoned.py`), validation (`backtest.py`, `select_backtest.py`, `effect_sizes.py`).
- **Product.** 2017–2020 gap-filled Td, 8,192 bucket files.
- **Figure and diagnostic scripts.** `reports/paper/scripts/`: QC, the three figure builders, and the warm-tail diagnostics, each reproducing specific numbers and figures in this manuscript.
- **Literature provenance.** `.asta/` holds the retrieved search and extraction records backing the citations added in Section 6, so each may be traced to its source record.
- **[TODO] Reproducibility gap.** `reports/results/` is excluded by `.gitignore` (line 64), so the Phase-A validation write-up is not under version control and should be committed before submission.
- **[TODO]** Deposit the product and a frozen code release; add DOIs.

9 AI-use disclosure

In accordance with CIP's Acceptable Use and Privacy policy: the analysis pipeline, validation harness, production fill driver and figures described here were developed and executed with AI assistance (Anthropic Claude, via Claude Code), under the direction and review of the author. The AI assistant wrote and ran analysis code, produced the figures, and drafted this manuscript. No confidential, personal, or unpublished third-party data was provided to the assistant; all inputs are institutional gridded climate products.

All quantitative results reported here are pending SME validation (Section 10). Numbers were read directly from the recorded outputs of the runs cited, not reproduced from memory.

10 SME sign-off

Table 5: Required sign-offs before submission.

Item	Reviewer	Date	Status
Validation design and baseline ladder (Section 4)			□
Hold-out results and effect sizes (Table 2, Table 3)			□
Seam-continuity test (Table 4)			□
Warm-tail disposition (Section 7)			□
Product release and metadata			□

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